

# SCD TYPE 1 LOGIC ON DELTA TABLE IN DATABRICKS

## Step 1:

I have uploaded my day 1 and day 2 csv files in my blob storage account.

The screenshot shows the Azure portal interface for a container named 'scdtype1'. On the left, the 'Overview' tab is selected, displaying the container's authentication method (Access key) and location (scdtype1 / 2025 / 03 / 30). A list of blobs is shown, including 'Day1.csv'. On the right, the '2025/03/30/Day1.csv' blob is selected, and its content is displayed in a table format. The table has 5 rows and 4 columns: ID, NAME, CITY, and PHONENUMBER.

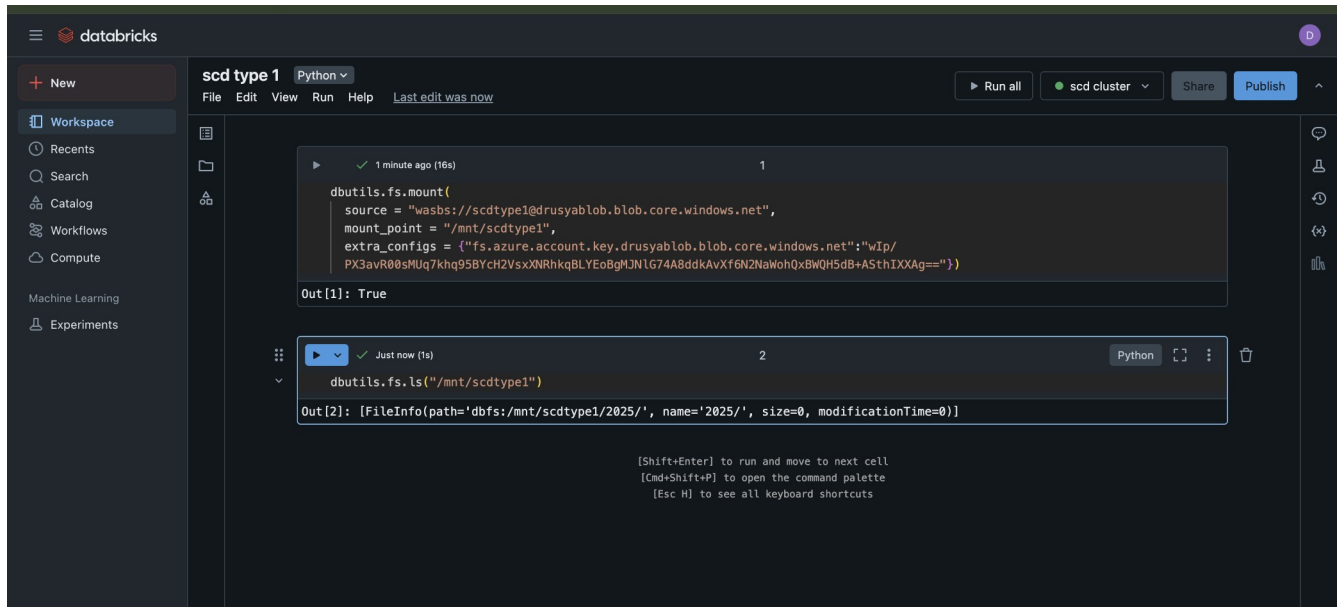
| ID | NAME   | CITY      | PHONENUMBER |
|----|--------|-----------|-------------|
| 1  | DRUSYA | KOTTAYAM  | 2200338812  |
| 2  | RIYA   | ERNAKULAM | 2345617890  |
| 3  | ARIYA  | CHEMMAI   | 3210985432  |
| 4  | PRIYA  | BLR       | 1092345672  |

The screenshot shows the Azure portal interface for the same container 'scdtype1'. On the left, the 'Overview' tab is selected, displaying the container's authentication method (Access key) and location (scdtype1 / 2025 / 03 / 31). A list of blobs is shown, including 'Day2.csv'. On the right, the '2025/03/31/Day2.csv' blob is selected, and its content is displayed in a table format. The table has 3 rows and 4 columns: ID, NAME, CITY, and PHONENUMBER.

| ID | NAME   | CITY       | PHONENUMBER |
|----|--------|------------|-------------|
| 1  | DRUSYA | TRIVANDRUM | 2200338812  |
| 3  | SANIYA | HYD        | 9901234231  |

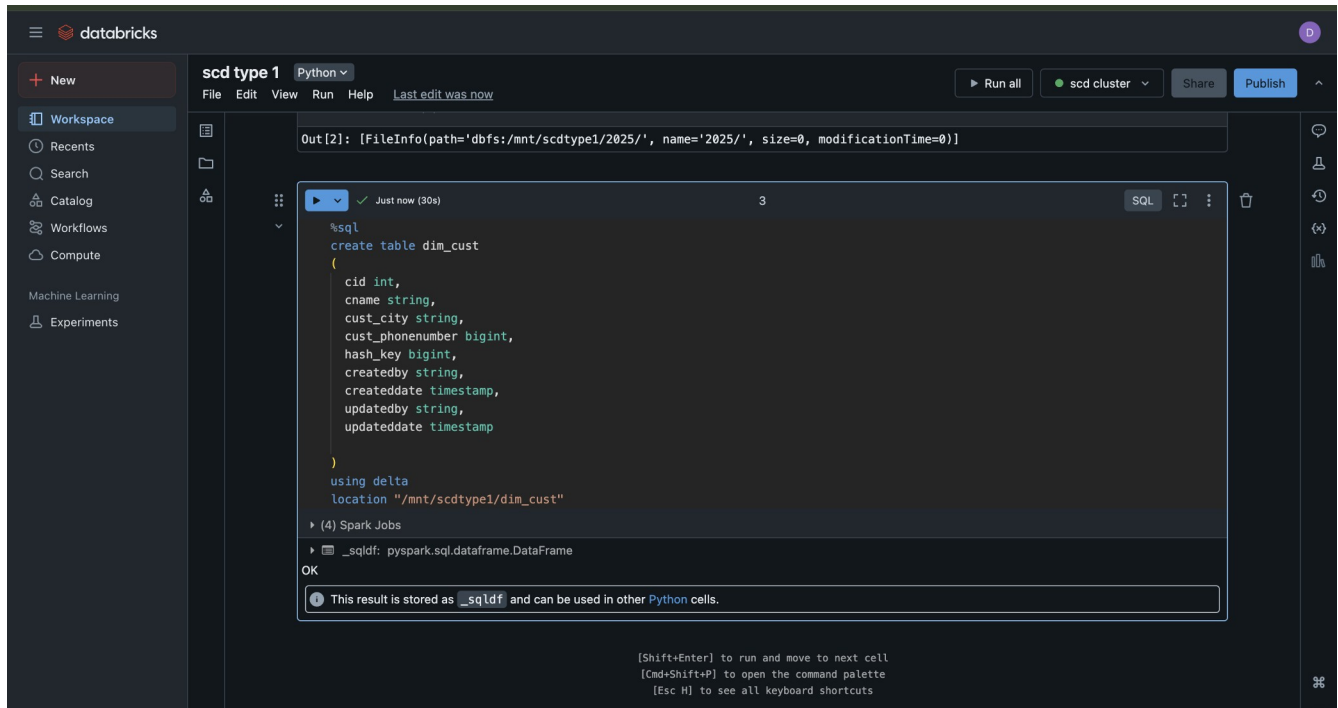
## Step 2:

Mounted my storage account using the access key to the databricks community edition my account.



## Step 3:

Creating a delta table in my account for storing the data.



Home > drusyablob | Containers >

**scdtype1** Container

Search

Upload Change access level Refresh Delete Change tier Acquire lease Break lease View snapshots Create snapshot Give feedback

**Overview**

Diagnose and solve problems

Access Control (IAM)

Settings

**Authentication method:** Access key (Switch to Microsoft Entra user account)

**Location:** scdtype1 / dim\_cust

Search blobs by prefix (case-sensitive)  ☐ Show deleted blobs

Add filter

| Name                                | Modified              | Access tier    | Archive status | Blob type  | Size | Lease state |
|-------------------------------------|-----------------------|----------------|----------------|------------|------|-------------|
| <input type="checkbox"/> [-]        |                       |                |                |            |      | ...         |
| <input type="checkbox"/> _delta_log |                       |                |                |            |      | -           |
| <input type="checkbox"/> _delta_log | 3/30/2025, 8:33:43 PM | Hot (Inferred) |                | Block blob | 0 B  | Available   |

## Step 4:

Created a widget as file date.

**scdtype1** Python

File Edit View Run Help Last edit was now

Run all scd cluster Share Publish

workspace

Recents

Search

Catalog

Workflows

Compute

Machine Learning

Experiments

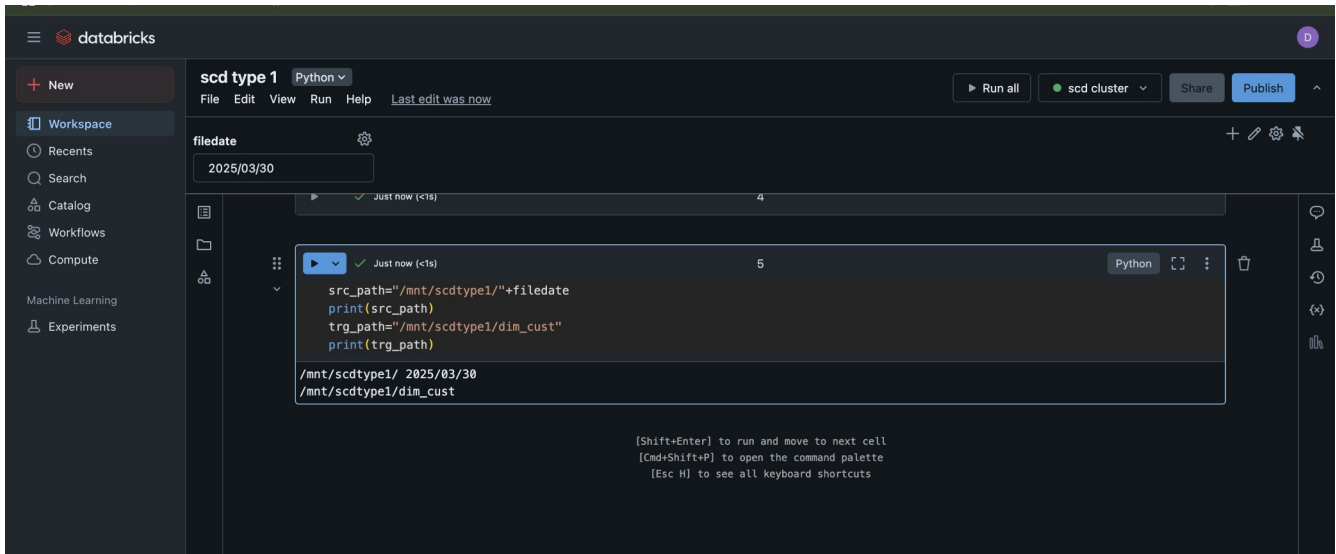
filedate

```
from pyspark.sql.functions import *
dbutils.widgets.text('filedate',' ')
filedate=dbutils.widgets.get('filedate')
```

[Shift+Enter] to run and move to next cell  
[Cmd+Shift+P] to open the command palette  
[Esc H] to see all keyboard shortcuts

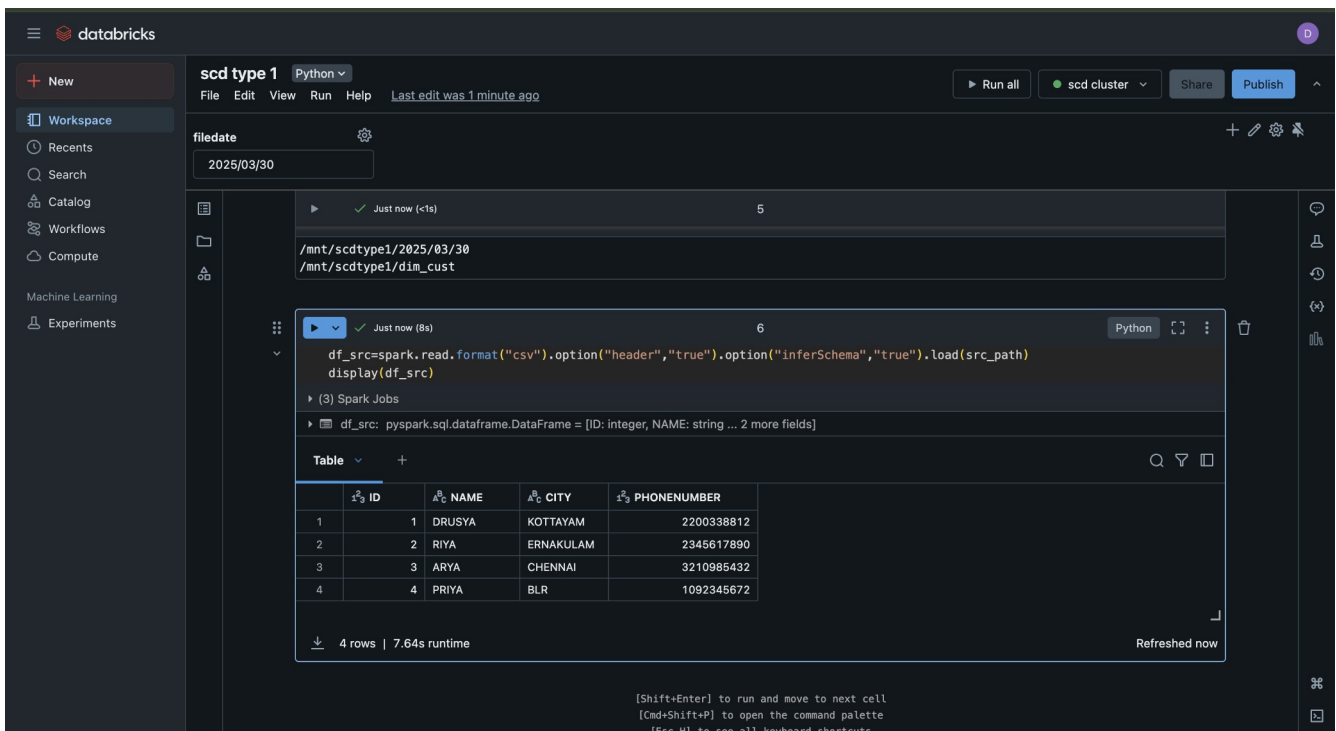
## Step 5:

Now I have entered my source path and target path using the file date that I created earlier.



## Step 6:

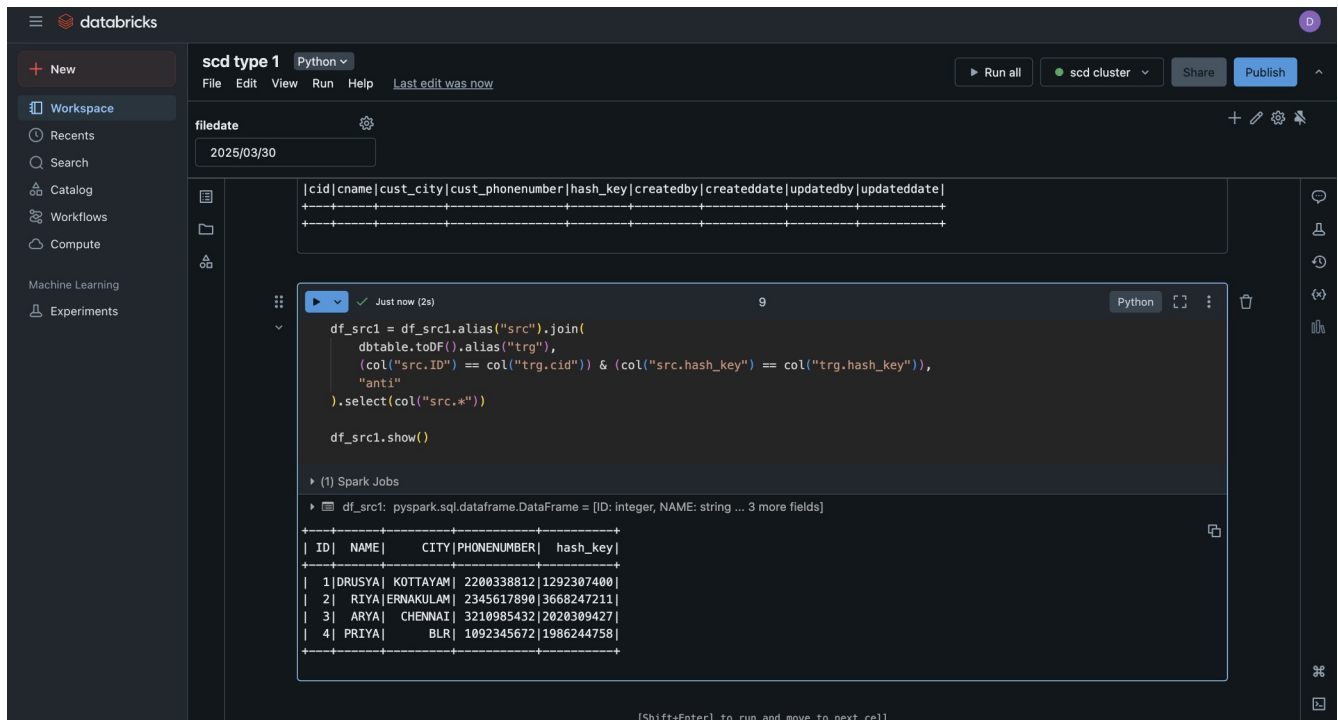
To display my source day 1 file I use the read command.





## Step 9:

Now I join my source file and target based on the ID so that in the next step I can compare the hashkey and ID to determine it is a insert or update.



The screenshot shows the Databricks workspace interface. The notebook is titled "scd type 1" and is in Python mode. The file date is set to 2025/03/30. The notebook cell contains the following Python code:

```
df_src1 = df_src1.alias("src").join(
    dbtable.toDF().alias("trg"),
    (col("src.ID") == col("trg.cid")) & (col("src.hash_key") == col("trg.hash_key")),
    "anti"
).select(col("src.*"))

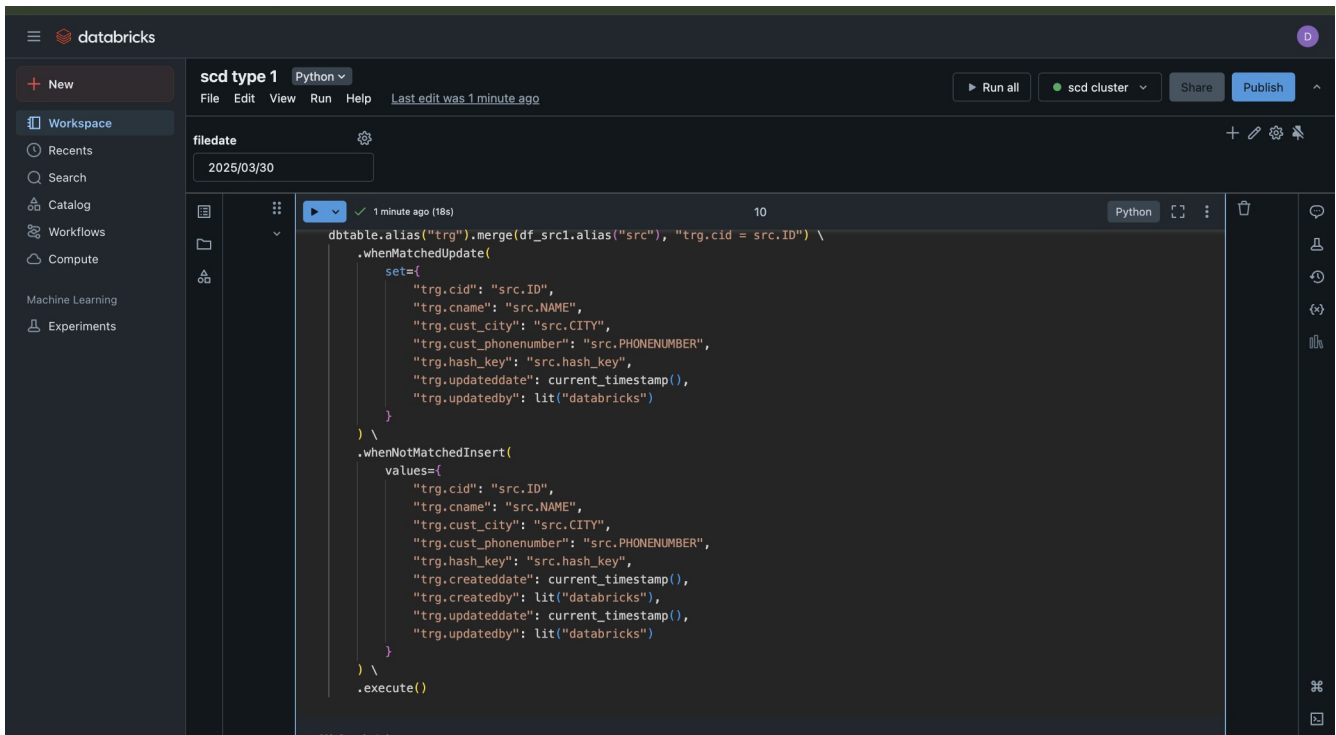
df_src1.show()
```

The output of the cell shows the Spark Jobs section with the following data:

| ID | NAME   | CITY      | PHONENUMBER | hash_key   |
|----|--------|-----------|-------------|------------|
| 1  | DRUSYA | KOTTAYAM  | 2200338812  | 1292307400 |
| 2  | RIYA   | ERNAKULAM | 2345617890  | 3668247211 |
| 3  | ARYA   | CHENNAI   | 3210985432  | 2020309427 |
| 4  | PRIYA  | BLR       | 1092345672  | 1986244758 |

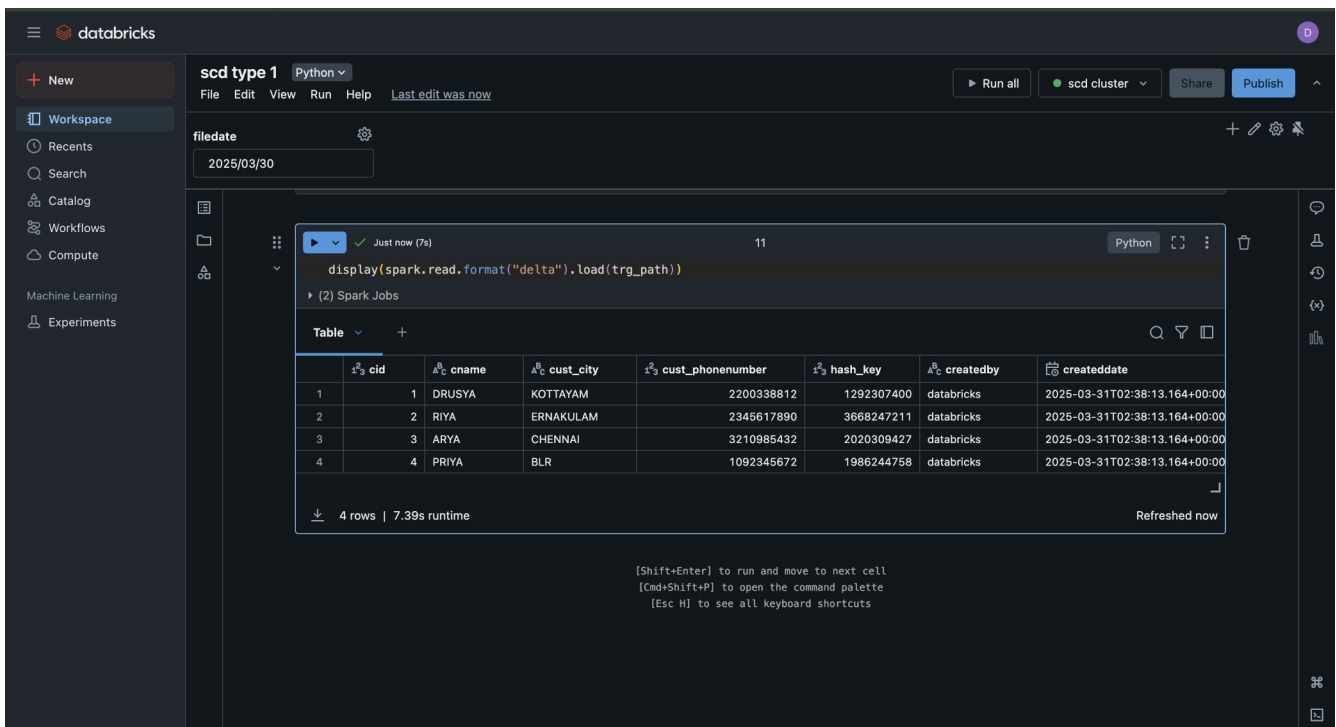
## Step 10:

Based on update or insert new records condition merge the files:



## Step 11:

Now reading the target file.



# Step 12:

Exit the notebook when success.

New

Workspace

Recents

Search

Catalog

Workflows

Compute

Machine Learning

Experiments

scd type 1

Python

File Edit View Run Help

Last edit was now

Run all

scd cluster

Share

Publish

filedate

2025/03/30

(2) Spark Jobs

Table

|   | cid | cname  | cust_city | cust_phonenumber | hash_key   | createdby  | createddate                   |
|---|-----|--------|-----------|------------------|------------|------------|-------------------------------|
| 1 | 1   | DRUSYA | KOTTAYAM  | 2200338812       | 1292307400 | databricks | 2025-03-31T02:38:13.164+00:00 |
| 2 | 2   | RIYA   | ERNAKULAM | 2345617890       | 3668247211 | databricks | 2025-03-31T02:38:13.164+00:00 |
| 3 | 3   | ARYA   | CHENNAI   | 3210985432       | 2020309427 | databricks | 2025-03-31T02:38:13.164+00:00 |
| 4 | 4   | PRIYA  | BLR       | 1092345672       | 1986244758 | databricks | 2025-03-31T02:38:13.164+00:00 |

4 rows | 7.39s runtime

Refreshed 1 minute ago

Just now (<1s)

12

Python

dbutils.notebook.exit("Success")

Notebook exited: Success

[Shift+Enter] to run and move to next cell

[Cmd+Shift+P] to open the command palette

[Esc H] to see all keyboard shortcuts

# Step 13:

Passing day 2 file and check the output.

New

Workspace

Recents

Search

Catalog

Workflows

Compute

Machine Learning

Experiments

scd type 1

Python

File Edit View Run Help

Last edit was 5 minutes ago

Run all

scd cluster

Share

Publish

filedate

2025/03/31

1 minute ago (23s)

10

(10) Spark Jobs

Just now (5s)

11

Python

display(spark.read.format("delta").load(trg\_path))

(3) Spark Jobs

Table

|   | cid | cname  | cust_city  | cust_phonenumber | hash_key   | createdby  | createddate                   |
|---|-----|--------|------------|------------------|------------|------------|-------------------------------|
| 1 | 2   | RIYA   | ERNAKULAM  | 2345617890       | 3668247211 | databricks | 2025-03-31T02:38:13.164+00:00 |
| 2 | 3   | ARYA   | CHENNAI    | 3210985432       | 2020309427 | databricks | 2025-03-31T02:38:13.164+00:00 |
| 3 | 4   | PRIYA  | BLR        | 1092345672       | 1986244758 | databricks | 2025-03-31T02:38:13.164+00:00 |
| 4 | 1   | DRUSYA | TRIVANDRUM | 2200338812       | 1810742831 | databricks | 2025-03-31T02:38:13.164+00:00 |
| 5 | 5   | SANIYA | HYD        | 9901234231       | 1390187022 | databricks | 2025-03-31T02:48:29.715+00:00 |

5 rows | 5.32s runtime

Refreshed now

Skipped

12

Python

dbutils.notebook.exit("Success")

Command skipped